MIL-V-21731D
22 June 1984
SUPERSEDING
MIL-V-21731C
4 August 1972

MILITARY SPECIFICATION

VIBRATORS, CONCRETE, ELECTRIC-MOTOR-DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lightweight, portable, double-insulated, electric-motor-driven concrete vibrators.

1.2 Classifications. Vibrators covered by this specification shall be one of the following types and styles, as specified (see 6.2):

Type I - Flexible shaft connected.

Style A - Swivel-based motor.
Style B - Hand-carried motor.

Type II - Motor in head.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

A-A-1586 - Tape, Pressure-Sensitive, Adhesive (Waterproof).
PPP-P-40 - Packaging and Packing of Handtools.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3895

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture and Fungus Resistant (for treatment of Communications, Electronic, and Associated Equipment).
MIL-T-22085 - Tape, Adhesive, Preservation and Sealing.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

C73-42 - Standards for Caps and Receptacles, Locking Type, 125 Volts, 15 Amperes, 2 Pole, 2 Wire Grounding Type, Specified Purpose.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D448 - Specification for Standard Sizes of Coarse Aggregate for Highway Construction.
D3951 - Commercial Packaging, Standard Practice For

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 45 - Portable Electric Tools.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1. Description. The type I concrete vibrator shall consist essentially of a sealed vibrating head connected to a double insulated electric motor by a flexible shaft. Type II vibrator shall be comprised of a sealed vibrating head with an integral double-insulated electric motor and a watertight hose encasing electrical cables. Type I, style A and B vibrators shall be furnished with quick disconnect type couplings to disconnect the flexible shaft from the electric motor. The illustrations of figure 1 are descriptive, but not restric- tive, and are not intended to preclude the purchase of vibrators otherwise conforming to the requirements of this specification.

3.2 First article. When specified (see 6.2), the contractor shall furnish a vibrator for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The vibrator shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the vibrators being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents
the latest production model.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Safety. Rotating mechanisms shall be protected and enclosed by rugged, waterproof casings. Particular care shall be taken in the design and construction of the vibrator to insure the safety of the operator from danger of electric shock.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means

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materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.7 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. When such contact cannot be avoided, joints between dissimilar metals shall be protected against galvanic corrosion by plating, coating, insulation, gaskets, or other suitable means. Bolts, nuts, pins, screws, and other fasteners shall be of the same material as the metals joined or shall be cathodic to the metals joined.

3.8 Design. Vibrators shall be of the heavy-duty type, simple in construction, reliable in operation, and all parts shall be readily accessible for inspection and adjustment.

3.9 Performance. The vibrator shall be capable of continuous operation without failure and, when tested as specified in 4.5.3, there shall be no evidence of moisture or dirt deposit inside the head, excessive wear, burnt commutator, worn or burnt brushes, or other defects detrimental to the vibrator.

3.10 Type I vibrators.

* 3.10.1 Motors. Type I vibrators shall be equipped with double insulated electric motors designed for intermittent duty and to operate in an ambient temperature of 40 degree Centigrade (°C). The motor for style A vibrator shall be totally enclosed, fan cooled, and shall operate on single-phase, 115 volts (V), alternating current (AC), 60 Hertz (Hz), electric power. Style B vibrator shall be furnished with an open, fan cooled splashproof, series wound, double insulated motor which shall operate on single-phase, 115 volts alternating current (VAC), 60 Hz electric power. When specified (see 6.2), a universal AC/DC (direct current), motor for style B vibrator which shall operate on single phase, 115 V, 60 Hz electric power, shall be furnished. All motors shall have horsepower (hp) ratings as specified (see 6.2) and equipment grounds permanently attached forming a continuous grounded circuit. Drive shaft shall be designed for positive attachment of the flexible shaft specified in 3.10.4. If brushes are used in the motors, they shall be the heavy-duty, self-adjusting type, and easily accessible for servicing and replacement.

3.10.2 Housings. Motor housing for style A vibrator shall have swivel base mount and integral carrying handle. Swivel base shall allow motor housing to rotate horizontally 360°. The style B vibrator motor housing shall incorporate a carrying handle, wraparound steel frame, and an eye for attachment of a shoulder strap. Style B vibrator shall be furnished with a shoulder strap.

3.10.3 Switches. A double pole, single throw switch shall be provided and mounted in or near the motor housing handle. Switch shall be furnished with a guard to prevent inadvertent actuation. Switch shall be capable of interrupting the stalled motor current at full rated voltage for 50 consecutive contacts at intervals of 10 seconds without the points sticking, burning, or flashing to the case.

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3.10.4 Flexible shaft. A flexible, steel wire shaft shall be used to transfer mechanical power from the motor drive shaft to the eccentric rotor in the vibrating head. The shaft shall be furnished in two sizes.

a. Short size shall be not less than 5 feet and not greater than 7 feet.
b. Long size shall be not less than 10 feet and not greater than 14 feet.

Lengths of shaft shall be as specified (see 6.2). The flexible shaft shall be designed for heavy-duty, high-speed service.

The flexible shaft shall be encased in a synthetic rubber casing that is impervious to the action of oil, acids, fluids, and abrasives expected to be encountered in concrete work. Case shall be fitted with threaded connectors on each end for connecting to motor and vibrating head. The threaded connectors shall be designed to allow free turning of the shaft and insure water tightness of the casing when coupled to the motor and vibrating head. Each vibrator shall be furnished with a ball bearing connector for coupling two flexible shafts.

3.10.5 Vibrating heads. Vibrating heads shall conform to the requirements specified in table I. The cap, shell, and base of the head shall be of abrasion-resistant steel or heat-treated steel. The head shall be completely sealed against the entrance of dust and moisture.

TABLE I. Type I vibrating head requirements.

Classification	Min frequency vibrations per minute	Cross-sectional area (sq in.) Min	Cross-sectional area (sq in.) Max	Length (in.) Min	Length (in.) Max	Volume (cu in.) Min	Volume (cu in.) Max
Type I, style A	8,500 (no load)	4.0	5.9	10	15	49	65
Type I, style B	9,500 (no load)	3.0	4.9	10	15	43	50

3.11 Type II vibrator.

* 3.11.1 Motor in vibrating head. Type II vibrator shall have the electric motor incorporated within the vibrating head. Motor shall be totally enclosed, nonventilated, designed for intermittent duty operation in an ambient temperature of 50°C. The motor shall be the universal type, double insulated, series wound, with a minimum 3/4 hp rating which shall operate on single-phase, 115 VAC, 60 Hz, electric power source. Motor shall have an equipment ground permanently attached forming a continuous grounded circuit. The eccentric rotor shall be an integral part of the ground circuit and the rotor shall be an integral part of the motor or shall be directly coupled to the motor. Vibrating head shall operate at a minimum frequency of 10,000 vibrations per minute. The vibrating head shall have a cross-sectional area of not less than 4.4 or more than 5.4 square inches, a length of not less than 15 or more than 18 inches, and a volume of not less than 70 or more than 92 cubic inches. The cap, shell, and base of the head shall be of abrasion-resistant steel and shall be completely sealed against the entrance of dust and moisture.

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3.11.2 Casing. Casing shall be furnished in two sizes, (A) short or (B) long, as specified (see 6.2). The short length casing shall be not less than 5 feet and not greater than 7 feet. Long size shall be not less than 10 feet and not more than 14 feet. Casing shall be of heavy-duty quality, lightweight, flexible, and constructed of not less than four plies of neoprene. The electrical power cable connecting the motor to the control switch shall be enclosed within the casing. Connectors used to secure the vibrating head and switch to the casing shall be a watertight joint.

3.11.3 Switches. The type II vibrator shall be equipped with thermal cutout relay and a waterproof control switch. The thermal cutout relay shall shut off the motor when overheated and automatically reset at a safe temperature.

* 3.12 Power cable. Each vibrator shall be furnished with a double insulated 3-wire power cable, with all wires having a size of not less than number 14 American Wire Gage. One end of the power cable shall be fitted with a locking type, 2-pole, 3-wire grounding type cap, conforming to ANSI C73.42. The oppo- site end of the power cable shall be secured to the vibrator control switch. The minimum length of the power cable for type I vibrators shall be not less than 2 feet and for type II not less than 25 feet. Requirements for power cables shall meet the applicable portions of UL 45.

3.13 Bearings. All rotating parts of the motor and vibrating heads shall be equipped with ball or roller bearings.

3.14 Lubrication. Vibrators shall be designed so lubrication is required not more than once every 40 hours of continuous operation, and serviceable in the field.

3.15 Threaded parts. All threaded parts shall conform to the applicable requirements of FED-STD-H28.

3.16 Radio-interference suppression. When specified (see 6.2), radio-interference suppression of the motor shall conform to current industry standards.

3.17 Fungus resistance. When specified (see 6.2), electrical connections, including terminal and circuit connections, components, and circuit elements shall be coated with varnish conforming to MIL-V-173 except that:

a. Components and elements inherently inert to fungi or hermetically sealed enclosures need not be treated.
b. Components and elements whose operation will be adversely affected by the application of varnish shall not be treated.

3.18 Carrying case. A portable carrying case, constructed of lightweight metal shall be provided with each vibrator. The case shall have ample capacity to house the motor, vibrating head, flexible shaft or casing, couplings, maintenance manual, repair parts, and tools for servicing the vibrator.

* 3.19 Identification and instruction plates. Each carrying case shall have an identification plate and an instruction plate affixed to it. Instruction plate shall contain clear, concise instructions for the operation of all controls, and shall describe all procedures to be followed in servicing the vibrator.

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3.20 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on vibrator when a first article is required (see 3.2, 6.2, and 6.4). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

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4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All vibrators of the same classification offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an acceptable quality level (AQL) of 2.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-2 and an AQL of 2.5 percent defective.

4.4 Examination. Each vibrator selected shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements shall constitute cause for rejection.

4.5 Tests. Each production vibrator selected shall be tested in accordance with 4.5.4. The first article vibrator shall be tested as indicated in paragraph 4.5.1 through 4.5.3. Failure of any vibrator to pass any specified test requirement shall constitute a defect.

4.5.1. Switches. The control switches and thermal cutout relay shall be tested to determine compliance with the requirements of 3.10.3 and 3.11.3, as applicable.

4.5.2 Radio-interference suppression. Conformance to the requirements of 3.16 shall be verified.

* 4.5.3 Performance test. Vibrator shall be operated continuously at full rated speed, at ambient temperature, for a period of 4 hours with the vibrating head submerged in water. At the completion of the 4-hour test, the vibrator shall be checked for lubrication, and oil or grease shall be replaced as required. Vibrator shall then be operated for an accumulated period of 36 hours over a duration of not less than 3 days. During this test, the vibrating head shall be immersed in a mixture of 250 parts of water, 1,250 parts of sand, 1,100 parts of number 2 gravel, and 800 parts of number 6 gravel. These designations are in accordance with ASTM D448. Every 4 hours, the vibrator shall be checked for excessive wear and heating. Performance requirements shall comply with 3.9. Vibration frequency shall be checked with a vibratory type tachometer to ascertain conformance to the requirements of table 1 and per paragraph 3.11.1. At the completion of the 40 hours of testing, the vibrator shall be disassembled and examined to confirm that there is no evidence of failure, and that the requirements of tests as referenced in 4.5 through 4.5.4 have had no deletorious effects on the vibrator.

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4.5.4 Operational test. Each vibrator selected shall be operated continuously for 1 hour in water. The control switch shall be operated as many times as necessary to determine ease of operation, effectiveness, and responsiveness, and that the motor and vibrating head actuated by the switch operate promptly, without restriction or malfunction. Bearings and shafts shall be examined for excessive heating or abnormal operation insofar as is practicable. Failure to meet the requirements of section 3 and any evidence of moisture inside the motor, vibrating head, shaft, or casing shall be cause for rejection.

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Unit protection. The complete vibrator shall be preserved method IIa. Openings in the equipment shall not be sealed, so that desiccated air may circulate. When a vibrator is furnished with a carrying case (see 3.18), desiccant shall be placed inside the carrying case.

5.1.1.6 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved with P-1 preservative.

5.1.1.7 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.8 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

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5.1.1.9 Consolidation. Disassembled components, tools, and publications for each vibrator shall be consolidated in containers. Contents of containers shall be cushioned, blocked, and braced to prevent movement in accordance with MIL-STD-1186.

5.1.2 Level B. Level B shall be the same as level A with the following modifications:

5.1.2.1 Unit protection. The complete vibrator shall be preserved method I or III as applicable. When a carrying case is furnished (see 3.18), the carrying case shall be utilized as the unit container when it meets the criteria of MIL-P-116.

5.1.2.2 Vibrator. All openings in the electric motor and controls shall be sealed with tape conforming to A-A-1586 or MIL-T-22085, type II.

5.1.3 Commercial. Material shall be packaged in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Packing shall be in accordance with MIL-STD-794. Only closed containers shall be selected from table I of MIL-STD-794.

5.2.2 Level B. Packing shall be in accordance with MIL-STD-794. When specified (see 6.2), only closed containers shall be selected from table I.

5.2.3 Commercial. Material shall be packed in accordance with ASTM D3951.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The vibrators covered by this specification are intended for use in consolidating low slump, portland cement concrete.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and style required (see 1.2).
c. Specifications and standards (see 2.1.1).
d. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
e. When a universal motor shall be furnished (see 3.10.1).
f. Horsepower ratings required (see 3.10.1).
g. Length of flexible shaft required (see 3.10.4).
h. Length of casing required (see 3.11.2).
i. When radio-interference suppression is required (see 3.16).
j. When fungus resistance is required (see 3.17).
k. Cleaning, treatment, and painting (see 3.20).
l. Level of preservation and level of packing required (see 5.1 and 5.2).
m. When closed containers are required (see 5.2.2).

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6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved (CDRL) (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required (see 3.2, 4.2.1, and 6.2), the item will be tested and should be a sample selected from the first vibrator production items or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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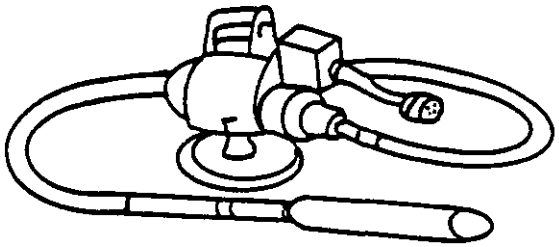
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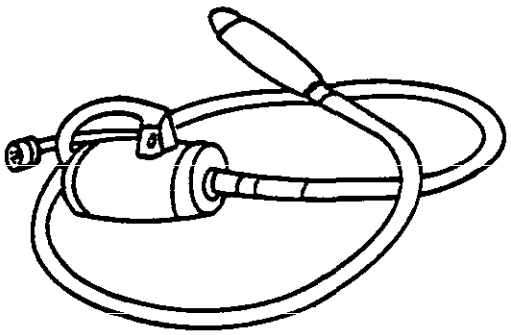
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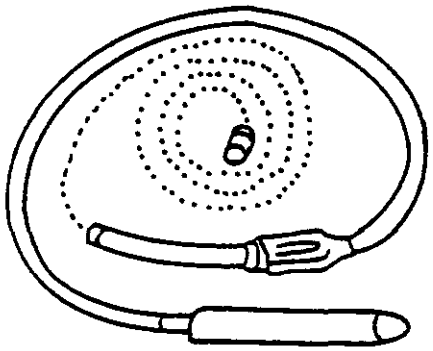


Type I - Flexible shaft connected.

Style A - Swivel based motor.



Style B - Hand carried motor.



Type II - Motor in head.

Figure 1. VIBRATORS, CONCRETE, ELECTRIC MOTOR DRIVEN

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-21731D	VIBRATORS, CONCRETE, ELECTRIC-MOTOR-DRIVEN

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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